# Sakthi Visahan Velayutha Moorthy

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## EDUCATION

| Institution | Degree / Board | Passing Year | Score / CGPA |
|---|---|---|---|
| Indian Institute of Technology Hyderabad | B.Tech | 2028 | 6.81 |
| THE VELAMMAL INTERNATIONAL SCHOOL | Class XII (CBSE) | 2024 | 90.80% |
| THE VELAMMAL INTERNATIONAL SCHOOL | Class X (CBSE) | 2022 | 94.00% |

## SKILLS

**Programming & Core Libraries:** Python, C, SQL, PyTorch, Scikit-Learn, Pandas, NumPy, SciPy, Matplotlib

**Machine Learning & Operators:** Generative AI, Physics-Informed Neural Networks (PINNs), Bayesian Neural Networks (BNNs), Fourier Neural Operators (FNO), Branch Net (DeepONet), XGBoost, Transfer Learning

**Mathematical Modeling & Stats:** Markov Chain Monte Carlo (MCMC), Metropolis-Hastings, Partial Differential Equations (PDEs), Finite Difference Method (FDM), Uncertainty Quantification

**Cloud, DevOps & Frameworks:** Git, CI/CD, REST APIs, Flask, PostgreSQL, GitHub Webhooks, OpenCV, Cloud Platforms

## PROJECTS

**Market Sentiment & Trend Analytics Predictor (NLP & Deep Learning)**

*Tech Stack: Python, SQL, Social Media APIs, Scikit-Learn, Deep Learning (LSTM)*

- **Market Trend Research:** Utilized live social media APIs and historical pricing data to research market trends and monitor fluctuations in consumer product popularity.
- **Actionable Data Synthesis:** Synthesized complex, qualitative behavioral data (public sentiment on product designs) and quantitative metrics into actionable goals for predictive forecasting.
- **Data-Driven Insights:** Applied machine learning to track how external upgrades or shifts in a brand's public perception directly impact market valuation, fulfilling a core need to translate complex data into business intelligence.

**GenAI-Powered Developer Analytics & CI/CD Pipeline**

*Tech Stack: Advanced SQL (PostgreSQL), Python, Google Gemini API, REST APIs*

- **Advanced SQL & Reporting:** Deployed and queried a relational PostgreSQL database to track historical developer commit data, utilizing Advanced SQL to monitor system performance over time.
- **Automated Performance Reporting:** Integrated the Google Gemini LLM to create real-time, automated reports on the performance and complexity of code submissions, suggesting possible improvements directly to stakeholders.
- **Stakeholder Collaboration:** Designed the pipeline to post actionable feedback directly back to GitHub threads, facilitating communication and driving product quality for engineering teams.

**Industrial IoT Performance Monitoring & Anomaly Analytics**

*Tech Stack: Python, XGBoost, Pandas, Data Analytics, SMOTE*

- **Product Performance Monitoring:** Monitored the operational performance of industrial conveyor belts by gathering and analyzing complex acoustic emission data.
- **Obsession to Simplify:** Synthesized massive, highly imbalanced datasets into a simplified, highly accurate predictive model, achieving an F1-score exceeding 92%.
- **Actionable Diagnostics:** Translated raw sensor data into clear, actionable diagnostic reports to identify rare, critical hardware failures and drastically reduce false-positive rates.

**Scalable Uncertainty Analytics & Risk Assessment Modeling (PI-BNN)**

*Tech Stack: Python, Probability & Statistics, Data Modeling*

- **Data-Driven Simplification:** Demonstrated an obsession to simplify complex technical systems by bypassing heavy computational mesh grids, slashing the required data samples from 20,000 to just 85 while maintaining output integrity.
- **Complex Data Synthesis:** Extracted dynamic uncertainty intervals from advanced mathematical models to create robust risk assessment pipelines.
- **Original & Productive Ideas:** Applied thermodynamics-inspired statistical sampling to overcome standard constraints, demonstrating critical, out-of-the-box thinking to solve analytical bottlenecks.

**LLM-Powered Strategic Decision Agent (2048 Game Environment)**

*Tech Stack: Python, Google Gemini Pro API, API Integration*

- **Strategic Analytics:** Built an automated agent that analyzes real-time grid states in the game 2048 to calculate and execute optimal strategic decisions.
- **Product Interaction Tracking:** Created automated feedback loops to monitor the AI's performance across different game states, generating data on success rates and strategic logic.

## RELEVANT COURSES

**Mathematics:** Calculus I, Calculus II, Elementary Linear Algebra, Differential Equations, Probability, Statistics, Transformation Techniques, Complex Variables.

**Computer Science & AI:** Introduction to Programming, Introduction to Deep Learning (MIT 6.S191 - Audited), Machine Learning (Stanford CS229 - Audited)

**Core Engineering:** Fluid Mechanics, Structural Analysis, Mechanics of Solids and Structures, Thermodynamics.

## EXTRACURRICULAR

**Athletics:** Represented IIT Hyderabad at the prestigious Inter-IIT Sports Meet '26 and secured medals at Mahindra University's Aero'26, successfully balancing rigorous academics with elite physical training.

**Football:** Core intra-college team member demonstrating strategic teamwork and rapid decision-making.

**National Service Scheme (NSS):** Actively volunteered for orphanage outreach, blood donations, and civic awareness programs. Mentored government school students on career paths and led campus sanitation drives.